22nd International Conference on Advanced Information Networking and Applications - Workshops

A Graphic Reputation Analysis System For Mining Japanese Weblog Based on both Unstructured and Structured Information

Guangwei Wang and Kenji Araki
Graduate School of Information Science and Technology, Hokkaido University
Kita 14, Nishi 9, Kita-ku, 060-0814 Sapporo, Japan
wgw,araki@media.eng.hokudai.ac.jp

Abstract

In today's rapidly expanding information environment, an immediate need exists for better tools to help a user dynamically navigate large result lists and search structures. However, at present a lot of research is focused only on the information analysis step, and is not continued on to the step that makes the analysis results visible, especially for Asian languages. In contrast, we propose a novel graphic reputation analysis system for Japanese. It not only integrates the analysis of unstructured opinion information and structured review rating data, but also presents the information using a simple and flexible GUI with visual mining graphs. The proposed system can help inexperienced users, who want to look over opinions on the Web before purchasing something, to make a quick decision. It also gives useful hints for other Asian languages. In this paper, we describe the detailed algorithms of the system, and show the performance through experiments.

1 Introduction

With the rapid growth of the Internet, people have an increasing number of opportunities to read reputations (personal opinions or beliefs), feelings (mental states) and sentiments (positive or negative judgments) that have been written or posted on web pages such as review sites, personal web sites, blogs or BBSes. Here, we use the term Weblog for these sites. Such subjective information can often be a useful basis for finding out what people think about a particular product, or for making a purchase decision. However, this reputation information is made up of an enormous amount of unstructured and/or semi-structured data which is very hard to manage in a short time.

A number of studies on automatically extracting and analyzing product reviews or reputations on the Web have been conducted [1, 2, 3, 4, 5]. These studies focus on using sentiment analysis to extract positive or negative information. Turney [1] works on opinion classification using word semantic orientation. He presents an unsupervised learning algorithm called SO-PMI (Semantic Orientation Using Pointwise Mutual Information) for classifying reviews as recommended or not recommended. Kobayashi et al. [2] define the main portions of an opinion as ⟨ object, attribute, opinion ⟩. They propose a technique for efficiently building an object name dictionary, an attribute expression dictionary, and an opinion word dictionary for any object domain. Popescu and Etzioni [5] introduced an unsupervised information-extraction system (OPINE) which mines reviews in order to build a model of important product features, their evaluation by reviewers and their relative quality across products. Most related work only focuses on the information analysis step, and does not continue on to the step that makes the analysis results visible. However, in today's continuing growth of personal opinions on the Web, there is a great need for visualizing tools to help users master and browse the information more easily.

In order to fulfill this need, we propose an original simple graphic reputation analysis system (called OMS-J) for Japanese Weblog reviews. The novelty of this system is two-fold: first, it combines the extraction and analysis of unstructured opinion information and structured review rating data from the Weblog. Secondly, this system presents an intuitive GUI using visual mining graphs aimed at inexperienced users who want to check opinions on the Weblog before purchasing something. The provided graphs can help a user to make a quick decision on which product is suitable with regards to some special feature (for example, design, price, battery, etc.). In the performance experiment, the system achieved over 60% top three ranking consistency, tested on 15 test products from three different categories (Electronic Dictionary, MP3, and Notebook PC) when compared to human rankings. This shows that the system ranks products similarly to humans. All evaluators showed an interest in using the system, since it is very time consuming and difficult to check reputation information on the Weblog manually. From the experiment results and the evaluators' feedback, we are convinced that in the future manual rating

978-0-7695-3096-3/08 $25.00 © 2008 IEEE
DOI 10.1109/WAINA.2008.75

IEEE computer society

evaluation can be supplemented by this kind of graphic reputation system to analyze and manage the information more effectively.

The remainder of this paper is organized as follows: Section 2 explains the background of this study. In Section 3, we describe the system in detail, including the algorithm of each process and the mining graphs GUI. Then Section 4 presents a comparative experiment to evaluate the performance of the system. Finally, Section 5 gives concluding remarks and directions for future work.

2 Background

2.1 Mining Technology

Web-based information repositories today are growing astronomically large - more than million terabytes of data. Within these huge amounts of data lies hidden information of strategic importance. Data mining, the extraction of hidden predictive information from large databases, is a powerful new technology with great potential to help users focus on the most important information in their data warehouses. Depending on the characteristics of the data, data mining can be divided into structured data mining and unstructured data mining. An algorithm for mining structured data is an algorithm that searches for interesting information in the given database structures. Unstructured data mining, on the other hand, is a cross between Information Retrieval (IR), Data Mining (DM), Artificial Intelligence (AI), and Natural Language Processing (NLP). It is also known as intelligent text analysis, text data mining or Knowledge Discovery in Text (KDT), and generally refers to the process of extracting information and knowledge, facts and meaningful relationships, from text documents. Opinion mining is a subdiscipline of text mining. The purpose of opinion mining is to mine reviews of various products by classifying them into positive or negative meaning from the "subjective" terms contained in the document. Automated opinion mining can provide a quick search and analysis result to both consumers and manufacturers [6].

2.2 Data Visualization

Both data mining and text mining use significant analytical methods and their results are often highly visual and graphical. Data visualization and information visualization techniques attempt to create an interface that is well suited for human decision making. Visualization and human-computer interactions (HCI) help to reveal conceptual associations and visualize knowledge maps [7].

3 Proposed System

3.1 System Outline

The proposed system aims to collect and analyze unstructured opinion information and structured review rating data on a specific type of product automatically from the Weblog, and then present the mining graphs of different types of this product category or the opinion distribution for a selected feature of the product. These graphs are multiform and can provide an easy intuitive visual comparison about each product on each feature. The system implementation of Weblog opinion mining and visualization can be seen in **Figure 1**. Steps 1 to 4 are repeated until the necessary information has been collected. In the next section, we describe the proposed system in detail. Step 1 and step 2 can be referred to in our previous study [11].

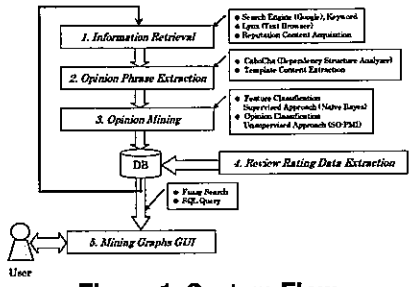


Figure 1. System Flow

3.2 Unstructured Data (Opinion) Mining

The opinion mining step performs feature classification and opinion classification using a supervised approach and an unsupervised approach respectively. The purpose of these classifications is to know what an opinion expresses about a certain product's features. This is the most important process in the system. The results of the classification are saved in a database and used to provide the mining GUI graphs.

3.2.1 Feature Classification

Feature classification aims to classify attribute expressions into feature categories (e.g. price, design, function, battery and neutral). Based on our previous study, we discovered that more than 90 percent of attribute expressions are nouns and different categories of products have similar attributes (such as "cost", "size" etc). This means that it is easy to create a feature training corpus. Therefore feature classification is performed by a supervised approach, a Naive Bayes Classifier.

3.2.2 Opinion Classification

Opinion classification classifies reputation expressions into positive, negative or neutral using an unsupervised approach, improved SO-PMI (Semantic Orientation Using Pointwise Mutual Information) [9]. The SO-PMI approach measures the similarity of pairs of words or phrases based on the mutual information theory, in our case the closeness of an opinion and words for "good" or "bad". The SO of a

phrase is calculated by Formula (1) which was presented in a previous study [9].

$$SO(P)=log_2\left[\frac{hits(P\ AND\ p_set)\ hits(n_set)}{hits(P\ AND\ n_set)\ hits(p_set)}\right]+0.8\cdot log_2\left[\frac{hits(p_set)}{hits(n_set)}\right] \quad (1)$$

P : *phrase*; "*p_set*"="*ii* (good) | *suki* (like) | *yoi* (good) | *miryoku* charm) | *tanosii* (delightful) | *daisuki* (favorite) | *yoku* (good) | *mannzoku* (satisfaction) | *omosiroi* (interesting) | *suteki* (lovely) | *uresii* (happy) | *yusyu* (interesting) | *subarasii* (excellent)"; "*n_set*"="*furyou* (poor) | *warui* (bad) | *fuan* (worry) | *muzukasii* (hard) | *ya* (dislike) | *kirai* (dislike) | *dame* (useless) | *turai* (painful) | *fukai* (dissatisfaction) | *fuman* (dissatisfaction) | *fuguai* (fault) | *saiaku* (worst) | *mazui*(bad)"

* The *Italic* part is Japanese.

3.3 Structured Data (Review Rating) Extraction

In most review sites, a five star review rating is also available in addition to the free review text input. Therefore, as addition information to the unstructured opinions, it is interesting to compare the review rating data too.

We extracted the review rating data on products from Kakaku.com[1]. "Kakaku.com" was chosen because it is the largest Japanese Weblog specializing in product comparisons of consumer goods, including price and user opinions, etc. A lot of people are giving reviews in the form of five star rankings. However, this site does not present a comparison on review rating data for each product in a visual form. In order to take advantage of the review rating data, we first locate the review page of Kakaku.com on the product, extract these rating data and save it to the database. Finally the ratings are compared and presented using intuitive graphs which are shown in **Section 3.4.4.**

3.4 Mining Graphs GUI

After the necessary information has been collected, the system provides the following four screens with mining graphs to show the opinion mining and information retrieval results in the database, based on our previous study [10].

3.4.1 The GUI for One Product Category

Figure 2 shows the distribution of positive and negative opinions of each feature type, such as "design", in four bar graphs. Two pie charts show the distribution of positive/negative opinions among the different product choices. Through these graphs, it is easy to quickly get an overview of the opinions of this category. This GUI is presented when a category name on the left list is selected (for example, "Electronic Dictionary").

3.4.2 The GUI for One Specific Product

This screen (**Figure 3**) shows the detailed mining results for one specific product, including the positive/negative opin-

[1] http://www.kakaku.com/

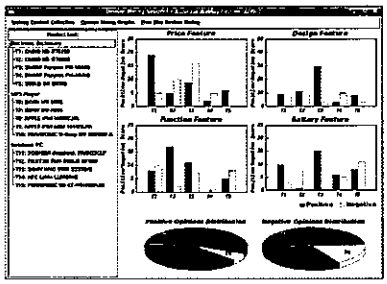


Figure 2. The GUI for One Product Category

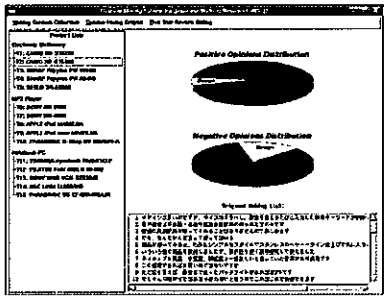


Figure 3. The GUI for One Specific Product

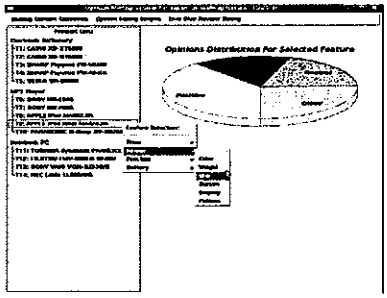


Figure 4. The GUI for One Product on Selected Feature

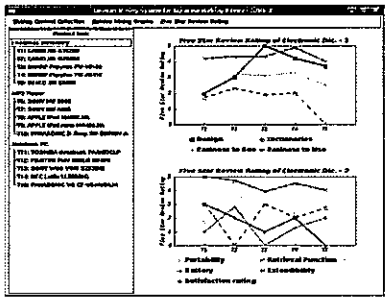


Figure 5. The GUI for Review Rating Data

ion distribution of each feature and the original Weblog contents list. Thus you can see the particular distribution for this product and also check the raw reputation information in textual format. This GUI is presented when the product name on the left list is selected (for example, "T2 CASIO XD-GT6800").

3.4.3 The GUI for One Product on Selected Feature

Furthermore, if you want to see the opinion distributions for one product on a specific feature, you can choose a feature in a pop-up menu for the selected product (**Figure 4**). In order to implement this function, we designed a Feature_Breakdown_Master table with the names of possible subfeatures. With the help of this table the subcategories of the feature, for example, "Design" can be broken down into "Color", "Weight", "Size", etc. For each subcategory, a list of related keywords and expressions is also available. Using fuzzy search and SQL queries, the relevant mining result can then be obtained from the database.

3.4.4 The GUI for Review Rating Data

This screen (**Figure 5**) presents the comparison of five star review ratings of all product types in the category in the form of graphs. Each curve denotes one feature (such as "Easiness to Use"). The last feature - Satisfaction rating shows the average of the rating data on all features. Thus from these graphs, not only can you see the individual rating comparisons for one particular feature, but you can also obtain an overall image of the products.

4 Performance Evaluation

Since part of the interface of the system is still being constructed, we only evaluated the opinion mining's performance. In this paper, we compared the results for 15 products to human rankings and to the rankings of an online sales site.

4.1 Experimental Setup

4.1.1 Test Products

There are the five products used for the three test categories-Electronic Dictionary, MP3, and Notebook PC. In the following experiment, the judgements of humans, an online sales site and our system are compared for these 15 products. These products were selected because they represent popular types from the rankings of Kakaku.com as of 2007.07.30.

4.1.2 Baseline

Human We asked three Japanese native speakers to check the reputation information of test products on the Weblog and deduce the ranking of each category on every feature (price, design, function and so on) manually. All of the evaluators are used to using the Web. For convenience of notation, E1, E2 and E3 are used in the following description to indicate the three evaluators.

Online Sales Site As mentioned in **Section 3.3**, Kakaku.com is an online sales site for comparisons of consumer goods. We collected the quantitative review ranking data for the test products from this site, and calculated the ranking of each feature. For convenience of notation, KA is used in the following description as a substitute for this site.

4.1.3 Our System

The evaluation scope of our system in these experiments is the information retrieval and opinion mining parts, excluding the five star rating data extraction.

4.2 Experimental Results

Table 1 presents the ranking results of each category and feature by the evaluators, Kakaku.com and the system for each of the 15 products (T1-T15). The data of KA was calculated and classified for each feature by the authors, according to the review and ranking data from Kakaku.com. The rankings of the system are calculated and sorted based on the SO value and classified feature category of each product in the database.

All the evaluators complained that it is very difficult to determine the ranking of each feature based on the Weblog reputation information, because this information is always disorderly and unsystematic. They often checked only a few favorite blogs to make the decision. Moreover, the ranking results are not very consistent between the three evaluators. It depends on the individual favorite and the retrieved pages.

Referring to the study of Hiroshima et al. [12] for evaluating the ranking results, we introduced the following consistency formulas to compare the ranking results of the system and the baselines (E1, E2, E3 and KA).

$$Top1 = \frac{\text{The number of Times } A\text{'s No.1 choices occur in } B\text{'s No.1 choices}}{\text{The number of } B \text{ choices}}$$

$$Top2 = \frac{\text{The number of Times } A\text{'s top 2 choices occur in } B\text{'s top 2 choices}}{\text{The number of } B \text{ choices}}$$

$$Top3 = \frac{\text{The number of Times } A\text{'s top 3 choices occur in } B\text{'s top 3 choices}}{\text{The number of } B \text{ choices}}$$

Top1 represents the No.1 ranking consistency between A and B. A is one system such as our system or E1 etc. B is the reference object, a set of baselines or all baselines. *Top2* is the top two ranking consistency. It represents the ratio of top two choices agreed upon by A and B, not considering the order. *Top3* is similar to *Top2*. Based on **Table 1** and the consistency formulas, we calculated the consistencies (*Top1*, *Top2*, *Top3*) between the baselines and the system for each category as shown in **Table 2**.

Table 1. The Ranking by Three Evaluators, Kakaku.com and System

Feature	By	Ranking (High→Low)		
		Electronic Dic.	MP3 Player	Notebook PC
Price	E1	T4,T3,T1,T2,T5	T10,T6,T7,T9,T8	T11,T12,T15,T14,T13
	E2	T4,T2,T3,T5,T1	T10,T9,T6,T7,T8	T11,T12,T14,T13,T15
	E3	T1,T2,T3,T5,T4	T8,T9,T10,T7,T6	T11,T14,T12,T13,T15
	KA	T4,T3,T1,T2,T5	T9,T7,T8,T6,T10	T12,T11,T14,T15,T13
	SYS	*T1,T3,T5,T2,T4*	*T6,T9,T7,T8,T10*	*T11,T14,T13,T12,T15*
Design	E1	T1,T2,T4,T3,T5	T9,T6,T10,T7,T8	T13,T14,T11,T12,T15
	E2	T3,T2,T1,T4,T5	T9,T10,T8,T6,T7	T14,T13,T12,T15,T11
	E3	T1,T2,T4,T3,T5	T9,T6,T8,T7,T10	T15,T13,T11,T12,T14
	KA	T1,T4,T3,T2,T5	T7,T8,T6,T9,T10	T14,T12,T13,T15,T11
	SYS	*T1,T2,T5,T3,T4*	*T9,T7,T6,T8,T10*	*T11,T13,T15,T12,T14*
Function	E1	T3,T4,T5,T1,T2	T6,T7,T10,T9,T8	T13,T15,T12,T11,T14
	E2	T2,T4,T3,T1,T5	T9,T8,T7,T10,T6	T15,T13,T14,T12,T11
	E3	T2,T1,T4,T3,T5	T9,T6,T8,T7,T10	T13,T15,T12,T14,T11
	KA	T5,T1,T4,T2,T3	T7,T9,T8,T10,T6	T14,T15,T13,T11,T12
	SYS	*T2,T1,T5,T3,T4*	*T9,T6,T7,T8,T10*	*T14,T13,T11,T15,T12*
Battery	E1	T1,T2,T5,T3,T4	T6,T7,T10,T9,T8	T15,T11,T14,T12,T13
	E2	T2,T1,T3,T4,T5	T10,T6,T9,T7,T8	T14,T11,T15,T12,T13
	E3	T4,T3,T2,T1,T5	T6,T10,T7,T9,T8	T15,T14,T13,T11,T12
	KA	T1,T3,T2,T4,T5	T7,T6,T10,T8,T9	T15,T14,T13,T12,T11
	SYS	*T3,T2,T1,T4,T5*	*T9,T6,T10,T8,T7*	*T14,T11,T13,T12,T15*
General Ranking	E1	T1,T2,T4,T3,T5	T6,T9,T10,T7,T8	T11,T13,T14,T12,T15
	E2	T2,T4,T1,T3,T5	T9,T10,T8,T6,T7	T14,T15,T13,T12,T11
	E3	T2,T1,T4,T3,T5	T9,T6,T8,T10,T7	T15,T13,T14,T12,T11
	KA	T4,T5,T2,T1,T3	T8,T9,T6,T7,T10	T11,T15,T13,T12,T14
	SYS	*T2,T1,T5,T3,T4*	*T9,T6,T7,T8,T10*	*T13,T14,T15,T12,T11*

E1: Evaluator1; E2: Evaluator2; E3: Evaluator3; KA: Kakaku.com; *SYS: System*

Table 2. The Consistencies between the System and the Baselines

%		ED	MP3	NPC	Avg.
Baseline1	*Top1*	40	35	25	33
(A = System,	*Top2*	55	58	50	54
B=$E1$,$E2$,$E3$,KA)	*Top3*	63	63	63	63
Baseline2	*Top1*	47	47	27	40
(A = System,	*Top2*	60	60	57	59
B=$E1$,$E2$,$E3$)	*Top3*	62	62	64	63

ED: Electronic Dictionary; NPC: Notebook PC

Table 2 shows the consistencies between the system and the baselines. Baseline1 includes all of the ranking results of the evaluators and Kakaku.com. On the other hand, baseline2 considers the results of the evaluators only. The average *Top1* values are under 40%, but all of the *Top3* values are over 60%. This means that the system presented similar top three ranking results as compared to the baselines. The results for Notebook PC are less consistent than the results for Electronic Dictionary and MP3. Furthermore, the consistency with baseline2 is better than the consistency with baseline1. That is because the data from Kakaku.com is based on the structured five star ranking data, not the unstructured reputation information. We consider this data to be different from the freely written opinions.

Table 3 presents the consistencies between the three evaluators by comparing E1 to E2*E3, E2 to E1*E3 and E3 to E1*E3. Each evaluator has a very similar *Top1*, *Top2* and *Top3*. The agreement between different annotators is quite low, showing that it is difficult and subjective to rank different products like this.

4.3 Discussion

Through this experiment, we found that manual rating evaluation, online sales sites and our system each have merits and demerits, which are summarized in **Table 4**. The demerits of human evaluation and online sales sites are usually difficult to overcome. On the other hand, the system has many merits and looks promising. After showing the outline and GUI of our system to the three evaluators, they

Table 3. The Consistencies of E1, E2 and E3

%		ED	MP3	NPC	Avg.
E1	*Top1*	20	40	40	33
(A = E1,	*Top2*	50	50	65	55
B = E2,E3)	*Top3*	70	70	73	71
E2	*Top1*	30	40	20	30
(A = E2,	*Top2*	50	50	65	55
B = E1,E3)	*Top3*	73	60	73	69
E3	*Top1*	30	50	40	40
(A = E3,	*Top2*	40	60	60	53
B = E1,E2)	*Top3*	70	77	77	70

ED: Electronic Dictionary; NPC: Notebook PC

were all very interested in our system. Although it is difficult to construct a knowledge system to analyze and present the reputation information automatically, and the system can only provide a rough analysis result, it still seems useful. We consider that it is necessary to continue to try and focus on how to build a highly precise and visual system.

Table 4. The Merits and Demerits of Human, Online Sales Site and System

	Merits	Demerits
Human	*Rational,Accurate*	*Time consuming, Impatient,Difficult*
OSS	*Simple,Instant*	*Inaccurate,Limited*
System	*Rapid,Easy to use, Flexible,External*	*Difficult to construct, Inexplicit,Rough analysis*

OSS: Online Sales Site

5 Conclusions

This study proposed an original graphic reputation analysis system for mining and presenting both unstructured opinion information and structured ranking data. In our previous study of opinion mining [9], we have presented our improved unsupervised SO-PMI algorithms and more than 70% precision for opinion sentence classification. Based on this algorithm, a Naive Bayes classifier, information retrieval and information visualization technology, we completed the framework of this system. In the experiments, we compared the ranking results for 15 test products of three categories for humans, an online sales site and our system. The system achieved 63% average consistency of the top three rankings compared to the human evaluators. This means that the system can conclude a similar judgement to humans.

With an intuitive interface using visual mining graphs and continuing to improve the analysis precision, we are convinced that this kind of system will be helpful to users and become a very strong tool to help in making purchasing decisions. It will also give useful hints for other Asian languages which have the similar characteristics (no space between words etc). In the future, we will continue to improve the classification precision and will try to evaluate the usability (GUI) and the entire performance of the system.

References

[1] Peter D. Turney, Thumbs up or thumbs down? Semantic orientation applied to unsupervised classification of reviews, Proceedings 40th Annual Meeting of the ACL, pp. 417-424, 2002.

[2] N. Kobayashi, K. Inui, Y. Matsumoto, K. Tateishi, and S. Fukushima, Collecting evaluative expressions by a text mining technique, IPSJ SIG NOTE, Vol.154, No.12, 2003, In Japanese.

[3] Taku Kudoh and Yuji Matsumoto, Applying Cascaded Chunking to Japanese Dependency Structure Analysis, Information Processing Society of Japan (IPSJ) Academic Journals, Vol 43, No 6, pp. 1834-1842, 2002, In Japanese.

[4] H. Yu and V. Hatzivassiloglou, "Towards Answering Opinion Questions: Separating Facts from Opinions and Identifying the Polarity of Opinion Sentences", Proc. EMNLP, Sapporo, Japan, pp.129-136, Jul. 2003.

[5] A. Popescu and O. Etzioni, "Extracting Product Features and Opinions from Reviews", Proc. HLT/EMNLP 2005, Vancouver, B.C., Canada, pp.339-346, Oct. 2005.

[6] K. Dave, S. Lawrence and D.M. Pennock, "Mining the Peanut Gallery: Opinion Extraction and Semantic Classification of Product Reviews", Proc. 12th International World Wide Web Conf., Hungary, pp.519-528, May 2003.

[7] Hsinchun Chen, Knowledge Management Systems-A Text Mining Perspective, Book, 2001.

[8] Computational Linguistics Laboratory, Graduate School of Information Science, Nara Institute of Science and Technology, CaboCha: A dependency structure analyzer

[9] Guangwei Wang and Kenji Araki, Modifying SO-PMI for Japanese Weblog Opinion Mining by Using a Balancing Factor and Detecting Neutral Expressions, Proc. NAACL HLT 2007, Rochester, NY, USA, pp 189-192, Apr. 2007.

[10] Guangwei Wang and Kenji Araki, OMS-J: An Opinion Mining System for Japanese Weblog Reviews Using a Combination of Supervised and Unsupervised Approaches, Proc. NAACL HLT 2007 Demonstration Program, Rochester, NY, USA, pp 19-20, Apr. 2007.

[11] Guangwei Wang and Kenji Araki, A Decision Support System Using Text Mining Technology, IEICE SIG Notes WI2-2006-6, pp.55-56, Jul. 2006.

[12] N. Hiroshima, O. Furuse, Y. Takahashi and R. Kataoka, Opinion Extraction and Summation System: [Opinion Reader], Proceedings of NLP2007 (The 13th annual conference), pp.432-435, 2007, In Japanese.